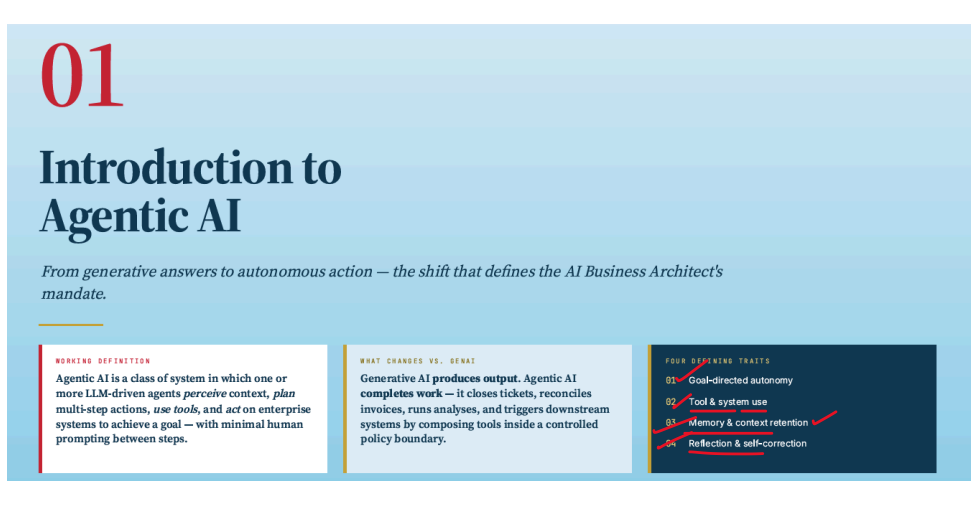
MS-AgenticAI-Architect



01

Introduction to Agentic AI

From generative answers to autonomous action — the shift that defines the AI Business Architect's mandate.

WORKING DEFINITION

Agentic AI is a class of system in which one or more LLM-driven agents *perceive* context, *plan* multi-step actions, *use tools*, and *act* on enterprise systems to achieve a goal — with minimal human prompting between steps.

WHAT CHANGES VS. GENAI

Generative AI **produces output**. Agentic AI **completes work** — it closes tickets, reconciles invoices, runs analyses, and triggers downstream systems by composing tools inside a controlled policy boundary.

FOUR DEFINING TRAITS

01 Goal-directed autonomy
02 Tool & system use
03 Memory & context retention
04 Reflection & self-correction

Agentic AI is an AI system where one or more AI agents can:

- Understand a goal
- Analyze the context
- Create a plan
- Use tools and software systems
- Execute actions
- Monitor results
- Adjust if needed
- all with minimal human intervention between steps.

Generative AI
"How do I process this invoice?"
- Then it:

Reads invoices
Validates supplier information
Checks purchase orders
Detects anomalies
Updates the ERP system
Generates a summary report

without requiring a human to guide every step.

* The Shift: Generative AI vs Agentic AI
Generative AI creates content.
Examples:
- Write an email
- Summarize a report
- Generate code
- Answer a question

Workflow: Human → AI → Output

- The human must still decide what actions to take.

Agentic AI
Agentic AI completes tasks.

Workflow

Human → Agent → Planning
↓
Tool Usage
↓
Business Action
↓
Outcome

--> Investigate declining sales.

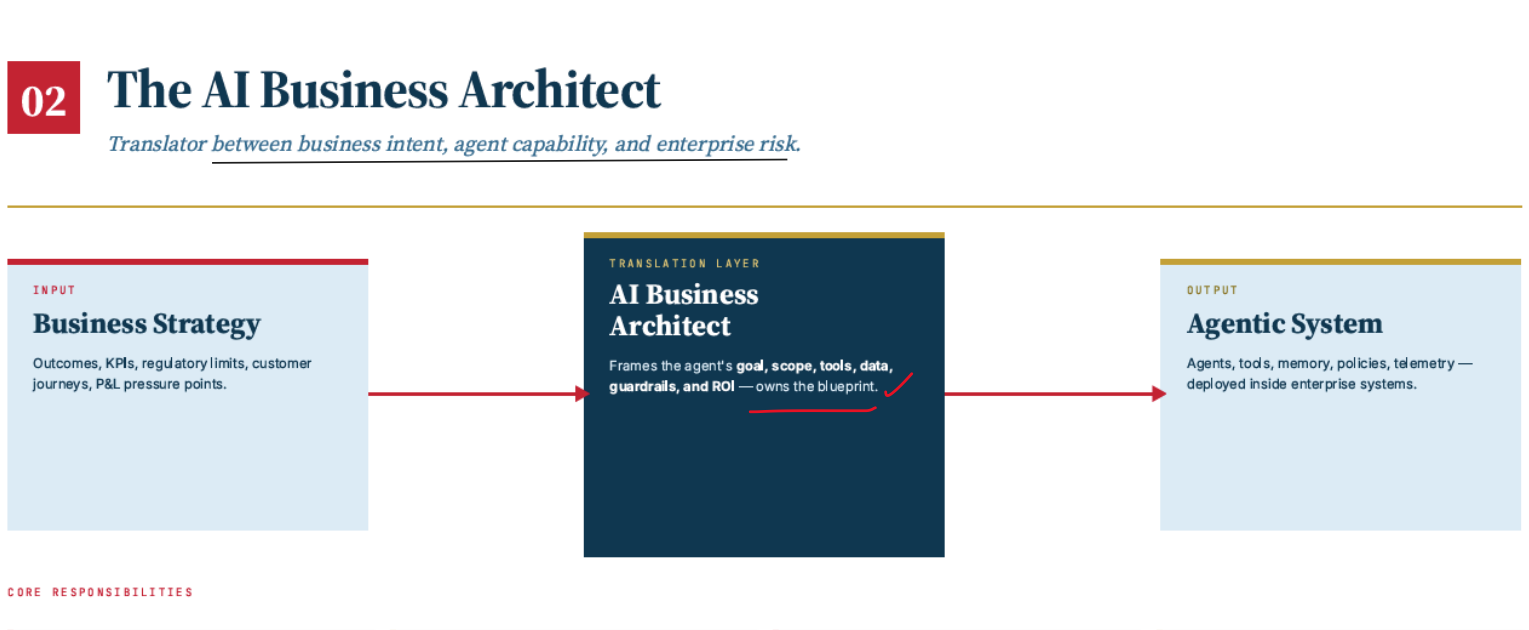
Agent:
- Retrieves sales data
- Identifies trends
- Finds root causes
- Creates report
- Alerts sales manager
- Opens follow-up tasks

The agent delivers a completed outcome, not just information.

Generative AI produces output. Agentic AI completes work.

This is the key distinction.

Generative AI	Agentic AI
Creates content	Achieves goals
Answers questions	Performs tasks
Single interaction	Multi-step workflow
Human executes	Agent executes
Provides advice	Takes action



02 The AI Business Architect

Translator between business intent, agent capability, and enterprise risk.

INPUT
Business Strategy
Outcomes, KPIs, regulatory limits, customer journeys, P&L pressure points.

TRANSLATION LAYER
AI Business Architect
Frames the agent's **goal, scope, tools, data, guardrails, and ROI** — owns the blueprint.

OUTPUT
Agentic System
Agents, tools, memory, policies, telemetry — deployed inside enterprise systems.

CORE RESPONSIBILITIES

R1 **Opportunity Framing**
Map business processes where autonomy beats automation; quantify time, $, risk.

R2 **Solution Blueprint**
Specify agent topology, tools, data contracts, human-in-the-loop checkpoints.

R3 **Governance & Trust**
Define policy, audit, escalation, and model-risk controls before production.

R4 **Value Realisation**
Tie agent telemetry to KPIs; close the loop from deployment to business outcome.

An AI Business Architect translates business goals into safe, measurable, and deployable AI agent systems.

Business leaders (who know the problem)
AI engineers (who build the solution)
Risk & compliance teams (who manage governance)

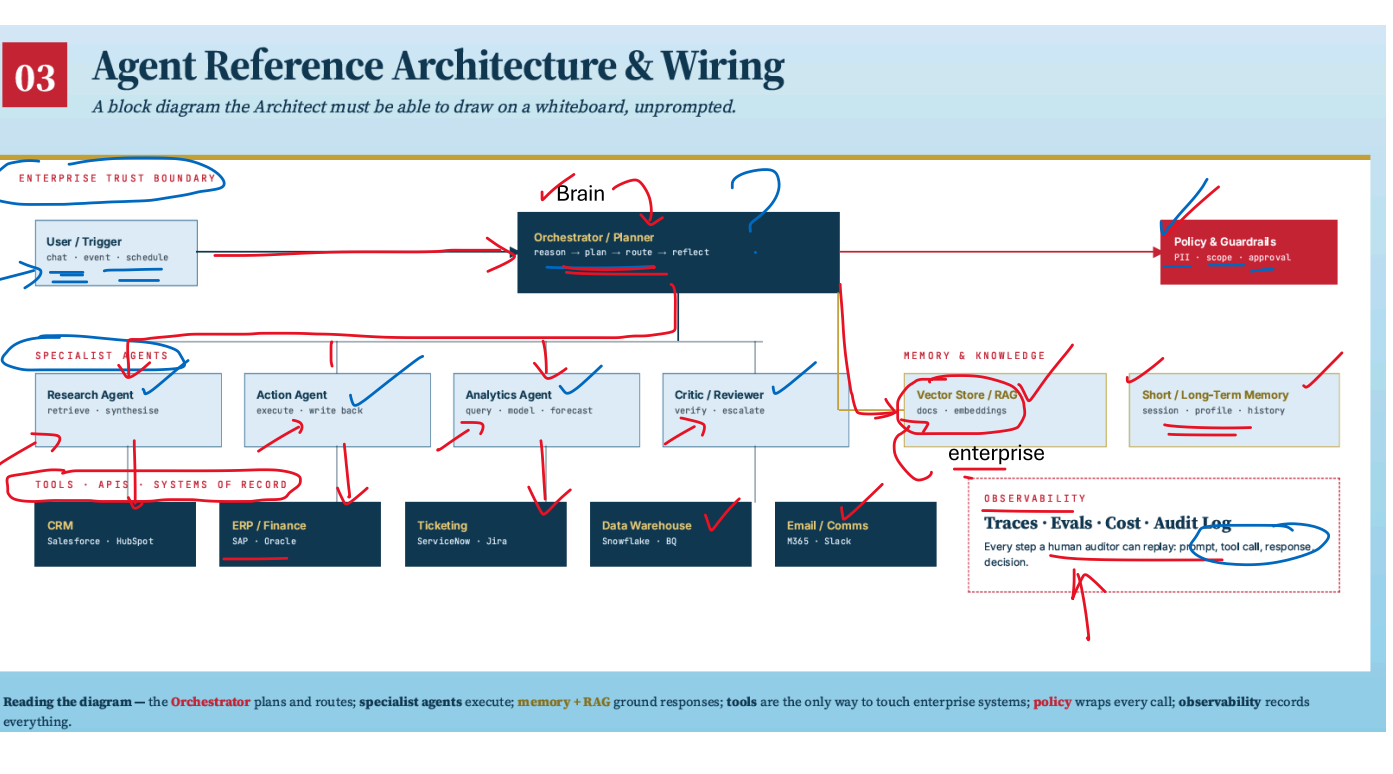
Business says: "We need a new office."
Engineers ask: "What should we build?"
Regulators ask: "Is it safe?"

The architect converts the vision into a blueprint.

AI Business Architect converts business strategy into an agentic AI system.

The Framework:

Business Strategy
↓
AI Business Architect
↓
Agentic System



03 Agent Reference Architecture & Wiring

A block diagram the Architect must be able to draw on a whiteboard, unprompted.

ENTERPRISE TRUST BOUNDARY

Brain

User / Trigger chat · event · schedule
Orchestrator / Planner reason → plan → route → reflect
Policy & Guardrails PII · scope · approval

SPECIALIST AGENTS

Research Agent retrieve · synthesise
Action Agent execute · write back
Analytics Agent query · model · forecast
Critic / Reviewer verify · escalate

MEMORY & KNOWLEDGE

Vector Store / RAG docs · embeddings
Short / Long-Term Memory session · profile · history

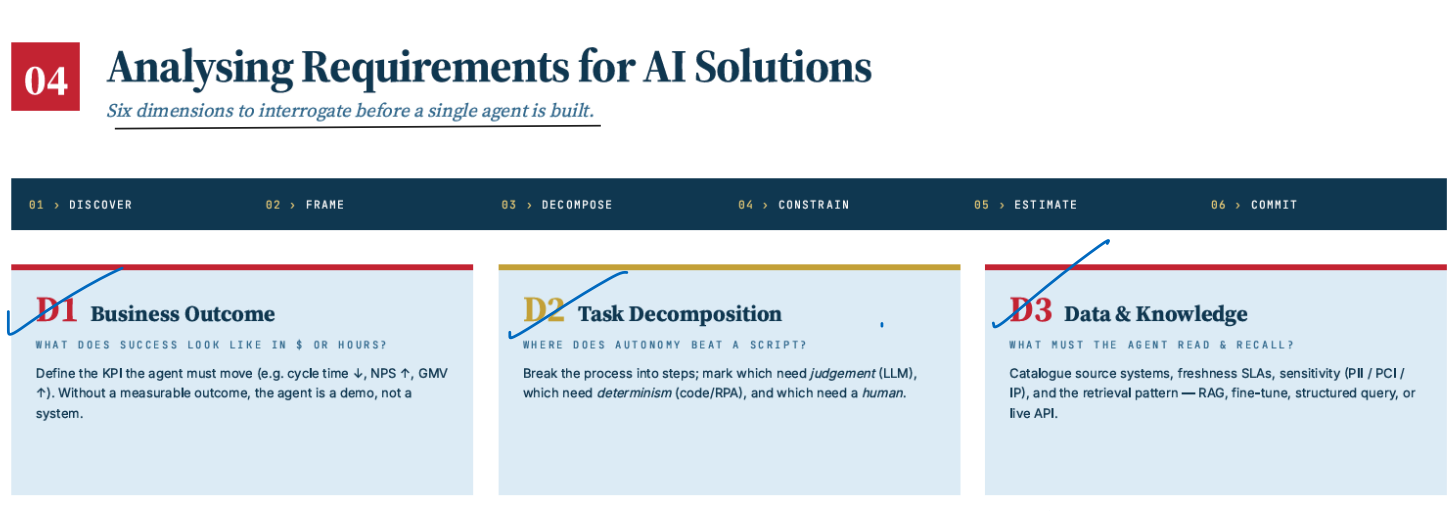
enterprise

TOOLS · APIS · SYSTEMS OF RECORD

CRM Salesforce · HubSpot
ERP / Finance SAP · Oracle
Ticketing ServiceNow · Jira
Data Warehouse Snowflake · BQ
Email / Comms M365 · Slack

OBSERVABILITY
Traces · Evals · Cost · Audit Log
Every step a human auditor can replay: prompt, tool call, response, decision.

Reading the diagram — the **Orchestrator** plans and routes; **specialist agents** execute; **memory + RAG** ground responses; **tools** are the only way to touch enterprise systems; **policy** wraps every call; **observability** records everything.



04 Analysing Requirements for AI Solutions

Six dimensions to interrogate before a single agent is built.

01 › DISCOVER 02 › FRAME 03 › DECOMPOSE 04 › CONSTRAIN 05 › ESTIMATE 06 › COMMIT

D1 Business Outcome
WHAT DOES SUCCESS LOOK LIKE IN $ OR HOURS?
Define the KPI the agent must move (e.g. cycle time ↓, NPS ↑, GMV ↑). Without a measurable outcome, the agent is a demo, not a system.

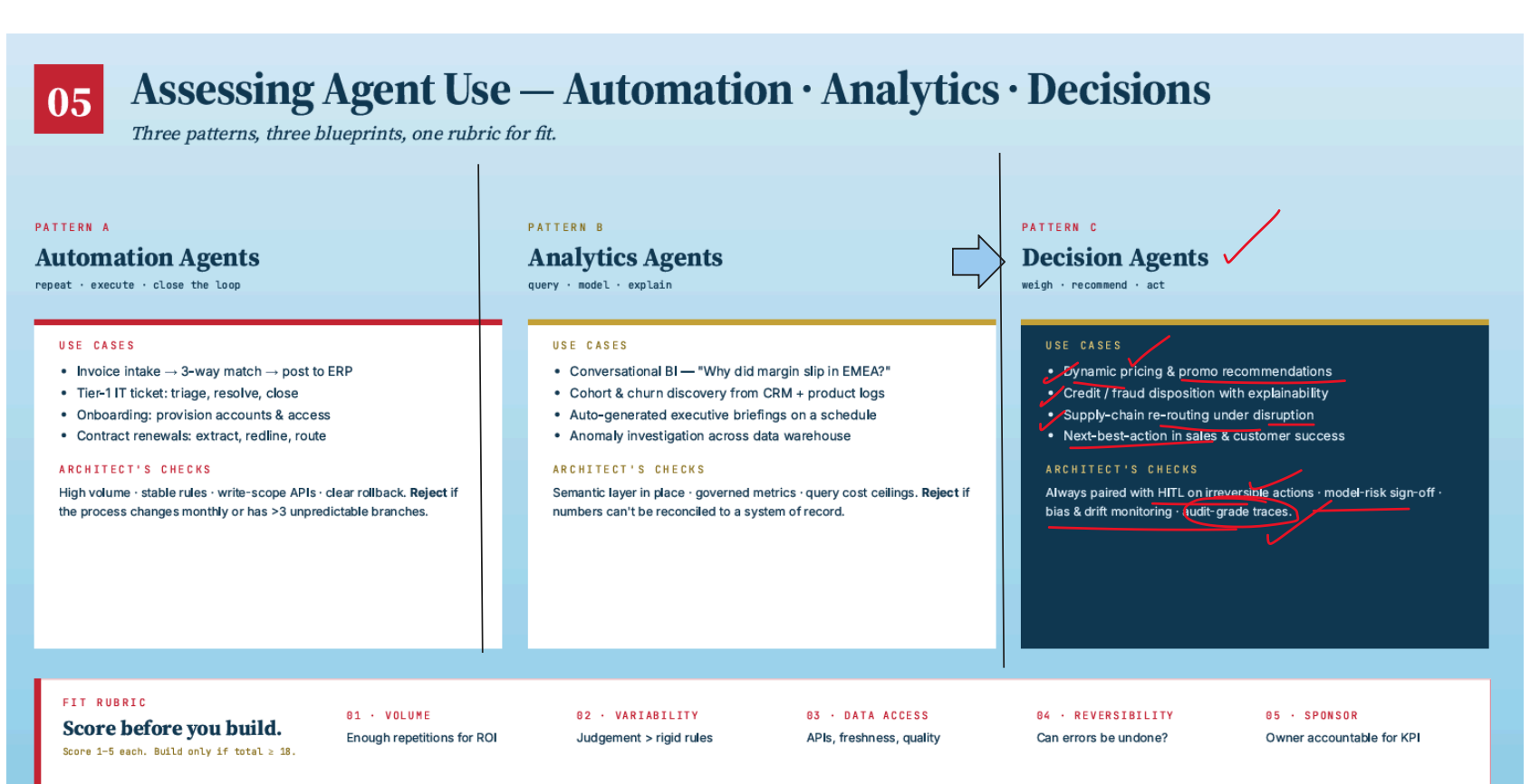
D2 Task Decomposition
WHERE DOES AUTONOMY BEAT A SCRIPT?
Break the process into steps; mark which need *judgement* (LLM), which need *determinism* (code/RPA), and which need a *human*.

D3 Data & Knowledge
WHAT MUST THE AGENT READ & RECALL?
Catalogue source systems, freshness SLAs, sensitivity (PII / PCI / IP), and the retrieval pattern — RAG, fine-tune, structured query, or live API.

D4 Tools & Integrations
WHAT CAN THE AGENT ACTUALLY DO?
Enumerate APIs, write-scopes, auth model, rate limits. An agent without tools is a chatbot.

D5 Risk & Guardrails
WHAT IS THE COST OF BEING WRONG?
Classify decisions by reversibility & blast radius. Set HITL gates, approval limits, and fallback paths accordingly.

D6 Economics & SLOs
WILL IT PAY BACK AT SCALE?
Model the unit economics — tokens per task × volume vs. labour saved — alongside latency, accuracy, and uptime SLOs.



05 Assessing Agent Use — Automation · Analytics · Decisions

Three patterns, three blueprints, one rubric for fit.

PATTERN A
Automation Agents
repeat · execute · close the loop

USE CASES
- Invoice intake → 3-way match → post to ERP
- Tier-1 IT ticket: triage, resolve, close
- Onboarding: provision accounts & access
- Contract renewals: extract, redline, route

ARCHITECT'S CHECKS
High volume · stable rules · write-scope APIs · clear rollback. **Reject** if the process changes monthly or has >3 unpredictable branches.

PATTERN B
Analytics Agents
query · model · explain

USE CASES
- Conversational BI — "Why did margin slip in EMEA?"
- Cohort & churn discovery from CRM + product logs
- Auto-generated executive briefings on a schedule
- Anomaly investigation across data warehouse

ARCHITECT'S CHECKS
Semantic layer in place · governed metrics · query cost ceilings. **Reject** if numbers can't be reconciled to a system of record.

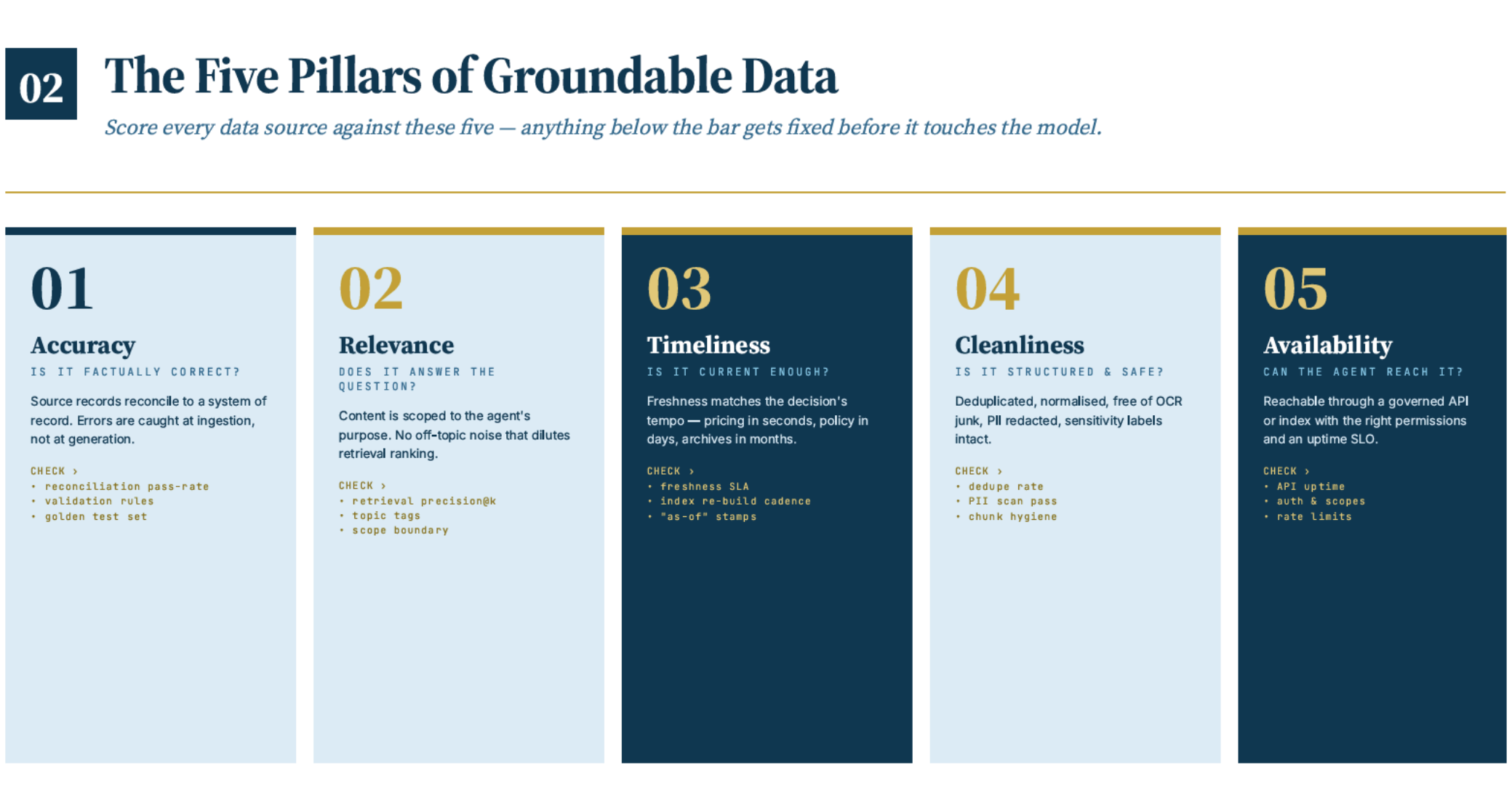
PATTERN C
Decision Agents
weigh · recommend · act

USE CASES
- Dynamic pricing & promo recommendations
- Credit / fraud disposition with explainability
- Supply-chain re-routing under disruption
- Next-best-action in sales & customer success

ARCHITECT'S CHECKS
Always paired with HITL on irreversible actions · model-risk sign-off · bias & drift monitoring · audit-grade traces.

FIT RUBRIC
Score before you build.
Score 1–5 each. Build only if total ≥ 18.

01 · VOLUME	02 · VARIABILITY	03 · DATA ACCESS	04 · REVERSIBILITY	05 · SPONSOR
Enough repetitions for ROI	Judgement > rigid rules	APIs, freshness, quality	Can errors be undone?	Owner accountable for KPI



02 The Five Pillars of Groundable Data

Score every data source against these five — anything below the bar gets fixed before it touches the model.

01 Accuracy
IS IT FACTUALLY CORRECT?
Source records reconcile to a system of record. Errors are caught at ingestion, not at generation.
CHECK ›
- reconciliation pass-rate
- validation rules
- golden test set

02 Relevance
DOES IT ANSWER THE QUESTION?
Content is scoped to the agent's purpose. No off-topic noise that dilutes retrieval ranking.
CHECK ›
- retrieval precision@k
- topic tags
- scope boundary

03 Timeliness
IS IT CURRENT ENOUGH?
Freshness matches the decision's tempo — pricing in seconds, policy in days, archives in months.
CHECK ›
- freshness SLA
- index re-build cadence
- "as-of" stamps

04 Cleanliness
IS IT STRUCTURED & SAFE?
Deduplicated, normalised, free of OCR junk, PII redacted, sensitivity labels intact.
CHECK ›
- dedupe rate
- PII scan pass
- chunk hygiene

05 Availability
CAN THE AGENT REACH IT?
Reachable through a governed API or index with the right permissions and an uptime SLO.
CHECK ›
- API uptime
- auth & scopes
- rate limits

Demos: Business Requirements & Data Readiness Assessment

Business case: ABC Corp HR team receives repeated employee questions on leave, WFH, payroll, and benefits.

Goal: -> Build an HRAgent - user self-service experience agent,

Copilot Studio Agent → Employee-facing conversational agent

Azure AI Foundry Agent → Backend reasoning / tools / enterprise workflow agent -

Part A: Capture Requirements for a Sample Process

Step 1: Start with Business Discovery

Before building any Copilot or Foundry agent, we must understand the business problem, success metrics, risk, and data readiness

Current HR support process:

Employees send HR queries by email.
HR manually searches policy documents.
HR replies after 12–24 hours. - (based on their SAL)

Step 2: Capture Business Requirements

Req ID	Requirement	Priority	Agent Mapping
BR-01	Answer HR policy questions instantly	High	Copilot Studio Agent
BR-02	Search approved HR documents only	High	Copilot knowledge / Foundry grounding
BR-03	Escalate complex cases to HR	High	Foundry Agent tool/action
BR-04	Provide source-backed answers	High	Azure AI Search / Foundry IQ
BR-05	Prevent salary/private data leakage	High	Guardrails + permissions
BR-06	Track usage and unanswered questions	Medium	Telemetry

Step-3: Which Agent Does What ?

Capability	Copilot Studio Agent	Azure AI Foundry Agent
Employee chat interface	Yes	Optional
FAQ-style HR answers	Yes	Yes
Generative answers from knowledge	Yes	Yes
Tool/API orchestration	Limited/Power Platform tools	Strong fit
Multi-step reasoning	Medium	Strong fit
Backend decision workflow	Medium	Strong fit
Enterprise monitoring/evaluation	Medium	Strong fit

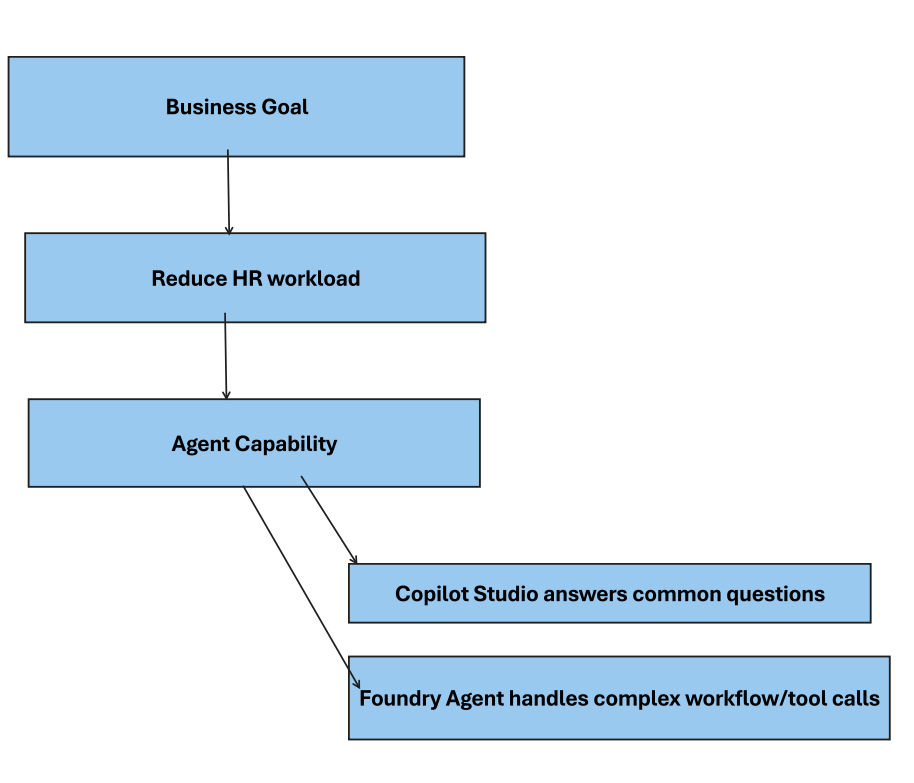
Step: 4 Define Current KPIs

KPI	Current Value
HR tickets/month	5,000
Avg response time	24 hours
SLA compliance	65%
Employee satisfaction	72%
Cost per ticket	₹75
Repeated queries	60%

Step: 5 Define Target KPIs

KPI	Current	Target
Avg response time	24 hours	2 minutes
SLA compliance	65%	95%
Employee satisfaction	72%	90%
Cost per ticket	₹75	₹20
Auto-resolution rate	0%	50–70%
Escalation accuracy	Manual	90%+ correct routing

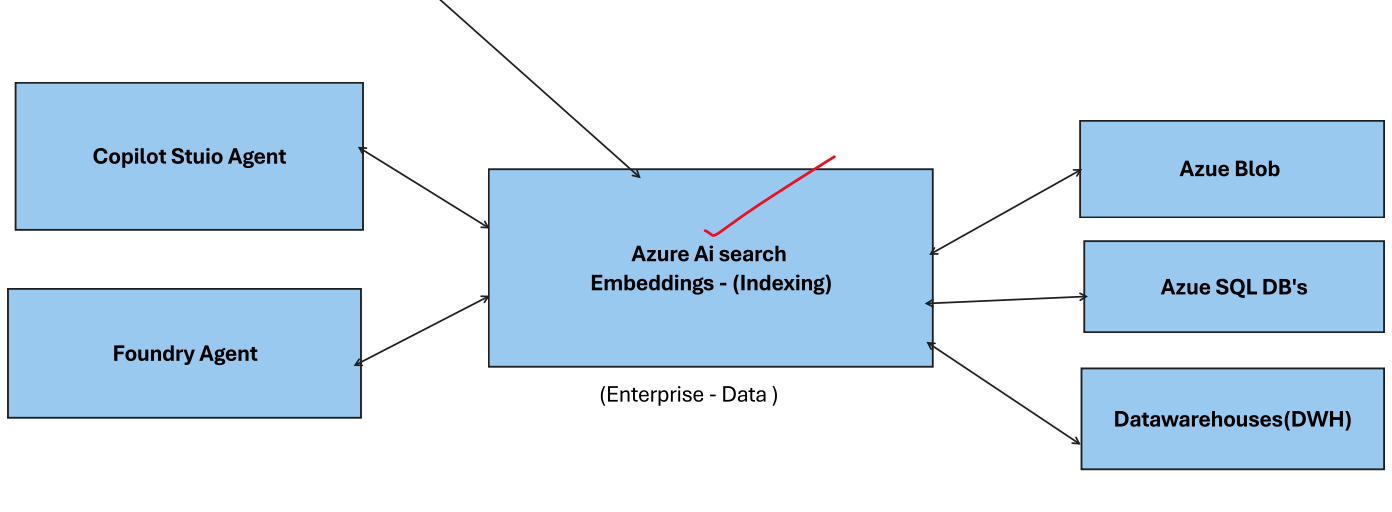
Step6: KPI-to-Agent Mapping

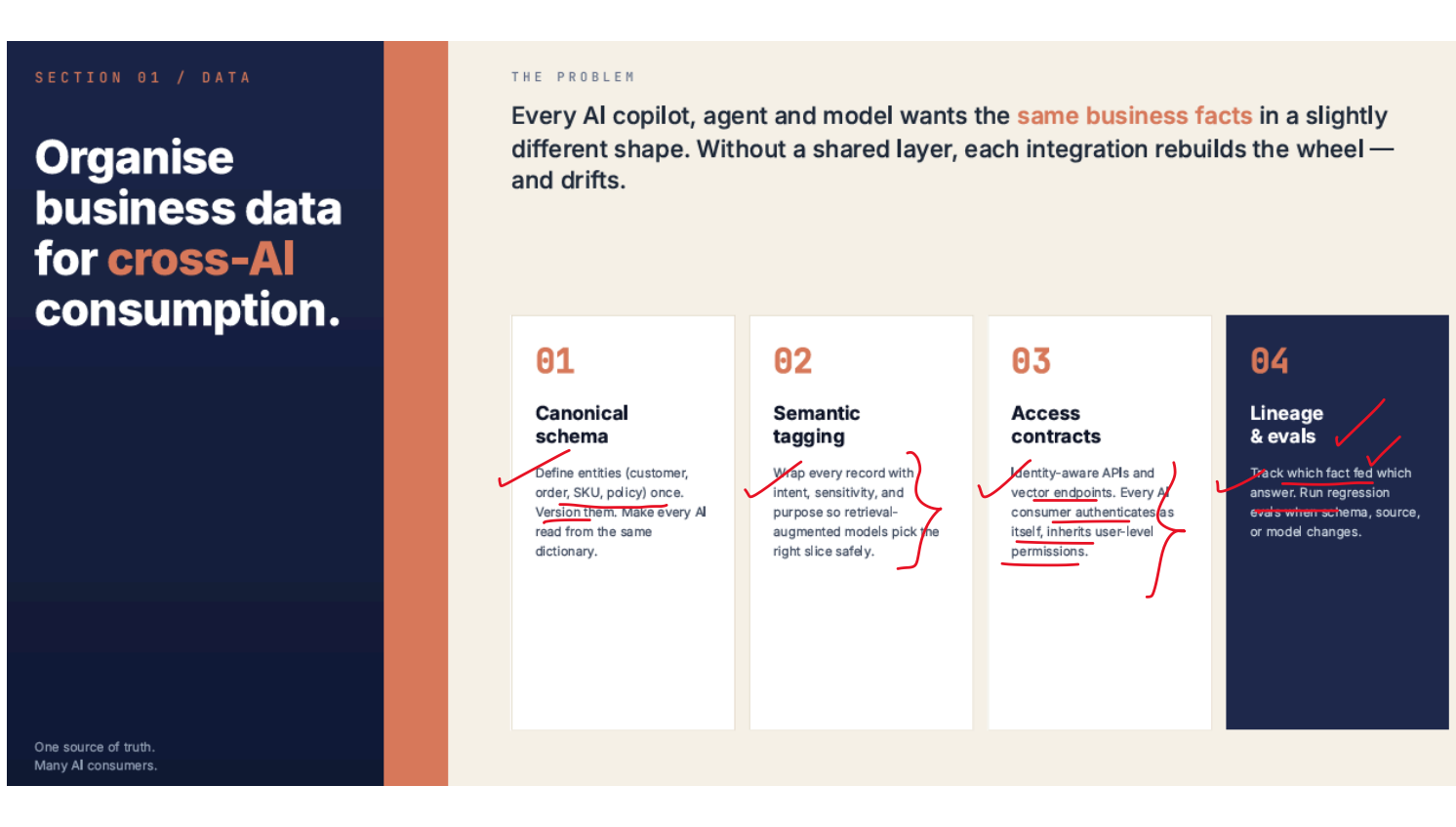

Business Goal
Reduce HR workload
Agent Capability
Copilot Studio answers common questions
Foundry Agent handles complex workflow/tool calls

- Provide Sample Datasets
(HR_Policy_2026.pdf
Leave_Policy.docx
Payroll_FAQ.xlsx
Employee_Master.csv
Benefits_FAQ.pdf)

-Explain Grounding Quality
Grounding means the agent uses trusted enterprise data instead of answering only from general model knowledge

- AI Search - connect Multiple Trusted Source like: Blob , SQL Database, Datawarehouses





SECTION 01 / DATA

Organise business data for cross-AI consumption.

One source of truth.
Many AI consumers.

THE PROBLEM

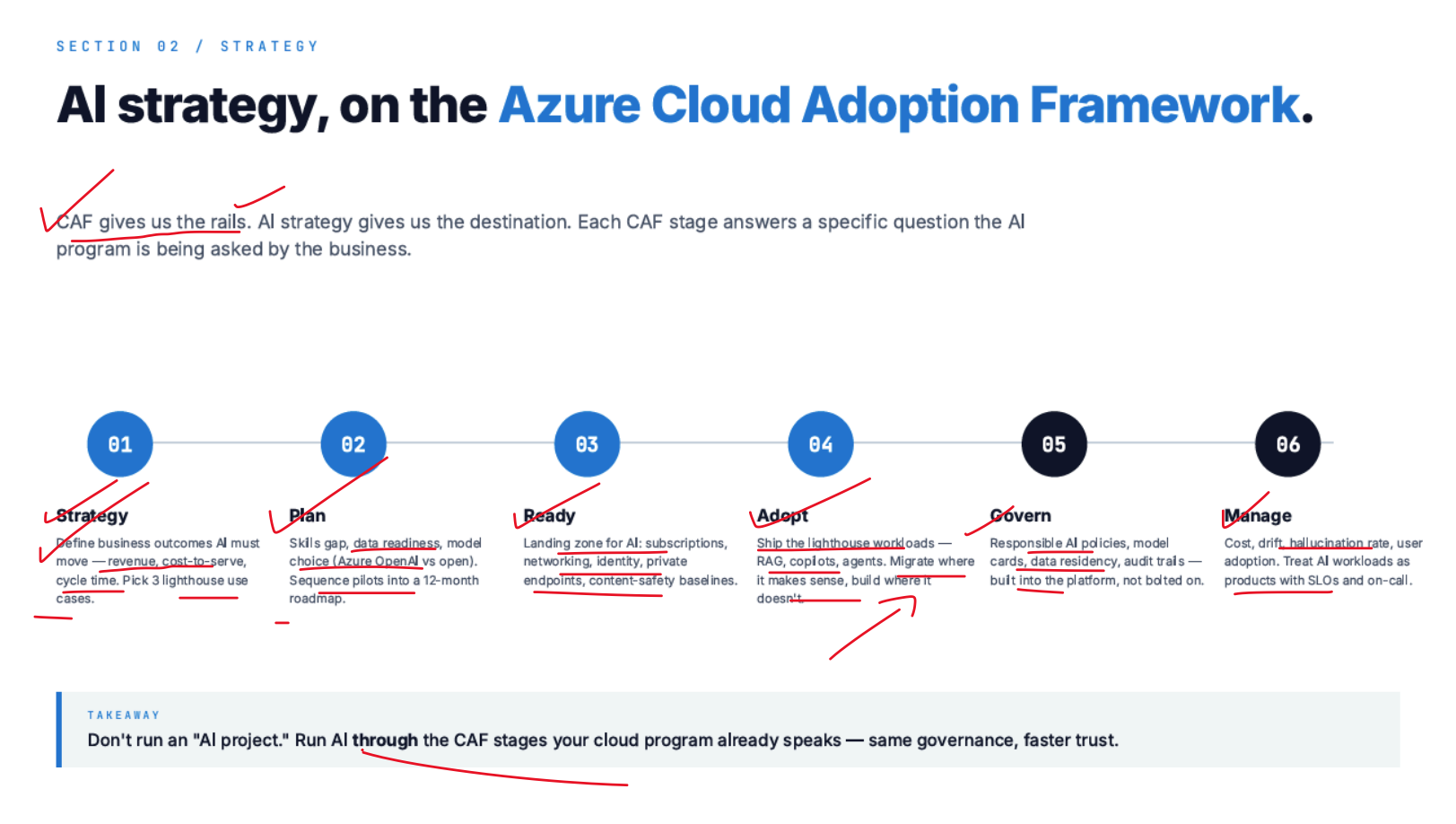
Every AI copilot, agent and model wants the same business facts in a slightly different shape. Without a shared layer, each integration rebuilds the wheel — and drifts.

01 Canonical schema
Define entities (customer, order, SKU, policy) once. Version them. Make every AI read from the same dictionary.

02 Semantic tagging
Wrap every record with intent, sensitivity, and purpose so retrieval-augmented models pick the right slice safely.

03 Access contracts
Identity-aware APIs and vector endpoints. Every AI consumer authenticates as itself, inherits user-level permissions.

04 Lineage & evals
Track which fact fed which answer. Run regression evals when schema, source, or model changes.



SECTION 02 / STRATEGY

AI strategy, on the Azure Cloud Adoption Framework.

CAF gives us the rails. AI strategy gives us the destination. Each CAF stage answers a specific question the AI program is being asked by the business.

01 Strategy
Define business outcomes AI must move — revenue, cost-to-serve, cycle time. Pick 3 lighthouse use cases.

02 Plan
Skills gap, data readiness, model choice (Azure OpenAI vs open). Sequence pilots into a 12-month roadmap.

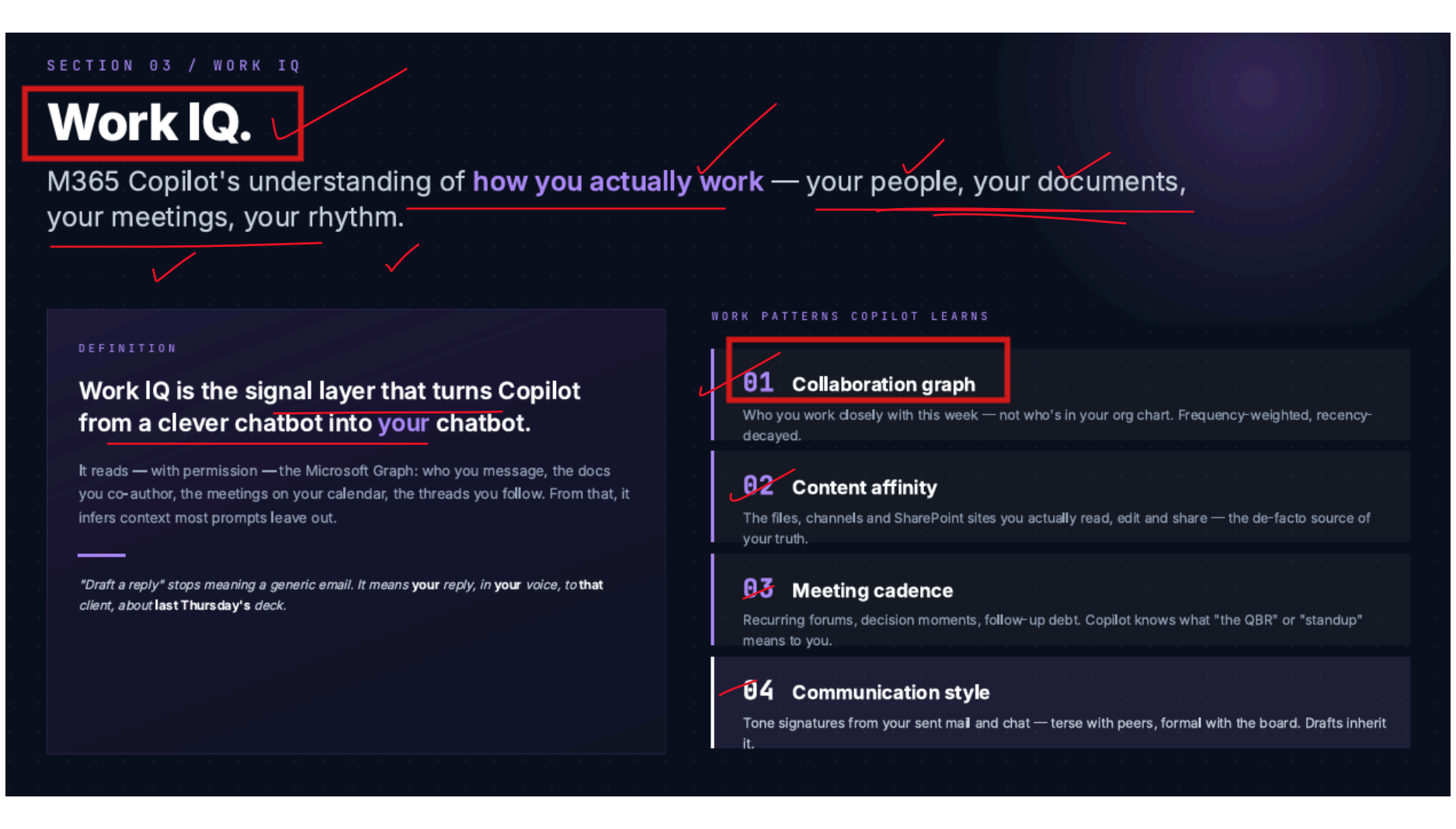
03 Ready
Landing zone for AI: subscriptions, networking, identity, private endpoints, content-safety baselines.

04 Adopt
Ship the lighthouse workloads — RAG, copilots, agents. Migrate where it makes sense, build where it doesn't.

05 Govern
Responsible AI policies, model cards, data residency, audit trails — built into the platform, not bolted on.

06 Manage
Cost, drift, hallucination rate, user adoption. Treat AI workloads as products with SLOs and on-call.

TAKEAWAY
Don't run an "AI project." Run AI **through** the CAF stages your cloud program already speaks — same governance, faster trust.



SECTION 03 / WORK IQ

Work IQ.

M365 Copilot's understanding of how you actually work — your people, your documents, your meetings, your rhythm.

DEFINITION

Work IQ is the signal layer that turns Copilot from a clever chatbot into your chatbot.

It reads — with permission — the Microsoft Graph: who you message, the docs you co-author, the meetings on your calendar, the threads you follow. From that, it infers context most prompts leave out.

*"Draft a reply" stops meaning a generic email. It means **your** reply, in **your** voice, to **that** client, about **last Thursday's** deck.*

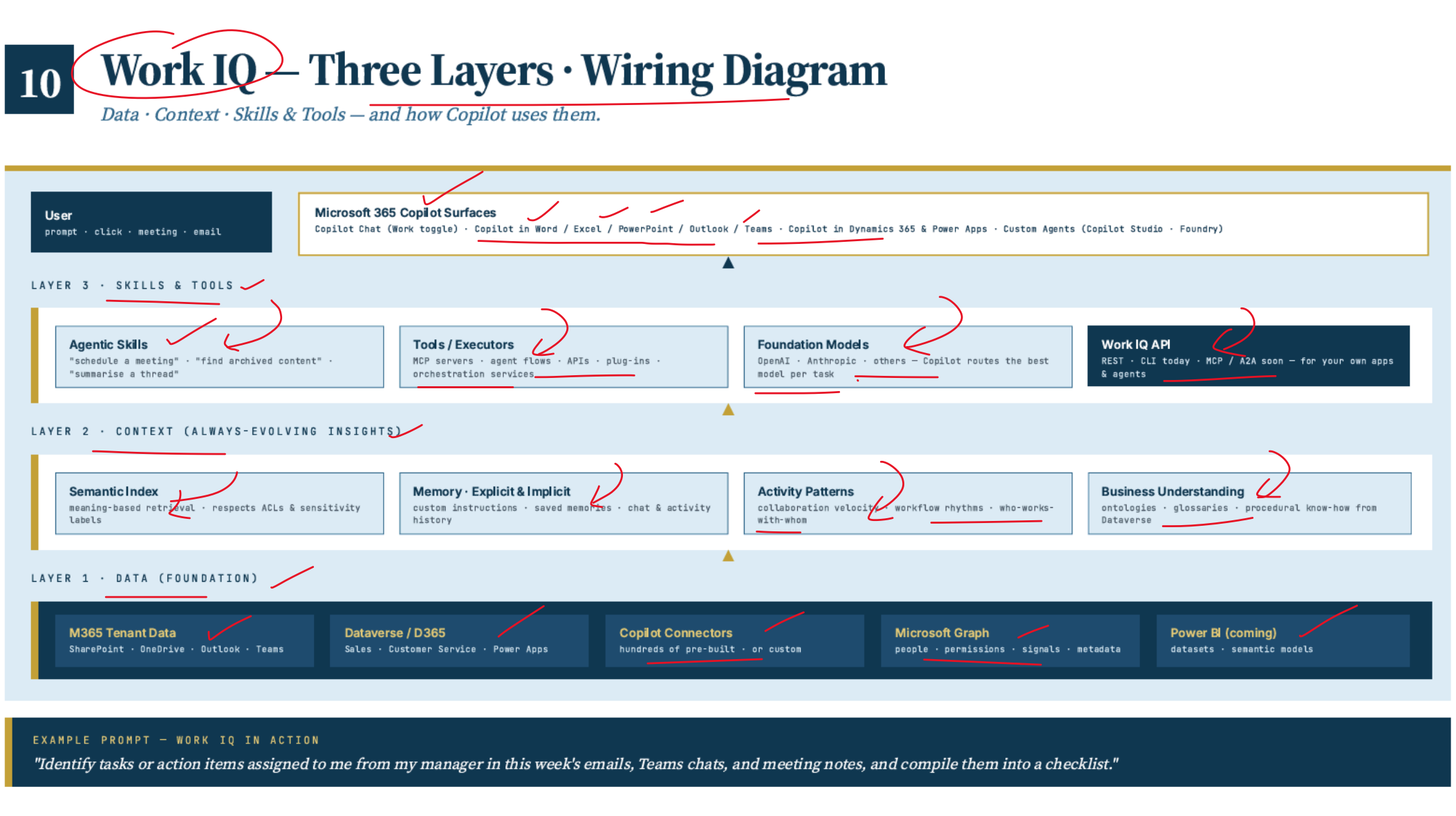
WORK PATTERNS COPILOT LEARNS

01 Collaboration graph
Who you work closely with this week — not who's in your org chart. Frequency-weighted, recency-decayed.

02 Content affinity
The files, channels and SharePoint sites you actually read, edit and share — the de-facto source of your truth.

03 Meeting cadence
Recurring forums, decision moments, follow-up debt. Copilot knows what "the QBR" or "standup" means to you.

04 Communication style
Tone signatures from your sent mail and chat — terse with peers, formal with the board. Drafts inherit it.



10 Work IQ — Three Layers · Wiring Diagram

Data · Context · Skills & Tools — and how Copilot uses them.

User
prompt · click · meeting · email

Microsoft 365 Copilot Surfaces
Copilot Chat (Work toggle) · Copilot in Word / Excel / PowerPoint / Outlook / Teams · Copilot in Dynamics 365 & Power Apps · Custom Agents (Copilot Studio · Foundry)

LAYER 3 · SKILLS & TOOLS

Agentic Skills "schedule a meeting" · "find archived content" · "summarise a thread"
Tools / Executors MCP servers · agent flows · APIs · plug-ins · orchestration services
Foundation Models OpenAI · Anthropic · others — Copilot routes the best model per task
Work IQ API REST · CLI today · MCP / A2A soon — for your own apps & agents

LAYER 2 · CONTEXT (ALWAYS-EVOLVING INSIGHTS)

Semantic Index meaning-based retrieval · respects ACLs & sensitivity labels
Memory · Explicit & Implicit custom instructions · saved memories · chat & activity history
Activity Patterns collaboration velocity · workflow rhythms · who-works-with-whom
Business Understanding ontologies · glossaries · procedural know-how from Dataverse

LAYER 1 · DATA (FOUNDATION)

M365 Tenant Data SharePoint · OneDrive · Outlook · Teams
Dataverse / D365 Sales · Customer Service · Power Apps
Copilot Connectors hundreds of pre-built · or custom
Microsoft Graph people · permissions · signals · metadata
Power BI (coming) datasets · semantic models

EXAMPLE PROMPT — WORK IQ IN ACTION
"Identify tasks or action items assigned to me from my manager in this week's emails, Teams chats, and meeting notes, and compile them into a checklist."